

## Part D — AI Usage Report

DSC 190/291 — Assignment 8 Student: Zeyu Bian

### D.1 Where I used AI

This assignment is entirely theoretical (Parts A–C are proofs; there is no experiment or code component), so AI played the role of a *proof collaborator and LaTeX/exposition assistant*, not a code debugger. Concretely I used AI to:

- **Stress-test the proof skeleton** I had drafted on paper for A.1–A.4 (the validation oracle inequality and its adaptive-tuning consequence), checking that each “ $\leq$ ” was justified and that I had not silently swapped  $n_T$  and  $n_V$ .
- **Check algebra** in the optimizations: the AM–GM balancing of  $\lambda\Psi(u) + \frac{2G^2}{\lambda\alpha n_T}$  (A.3), the  $c + 1/c$  discretization constant, the  $n_T = n_V = n/2$  simplification (A.4), the  $\lambda$  choice in B.4/B.5, and the Cauchy–Schwarz step in C.3.
- **Improve exposition**, turning my margin notes into clean markdown with consistent notation.

### D.2 Suggestions I accepted

1. **The “expectation of the max” route for A.1.** I initially planned a high-probability bound (union bound + Hoeffding tail, then integrate). AI suggested going directly through the subgaussian maximal inequality  $\mathbb{E} \max \leq \sigma\sqrt{2\log(2K)}$  with  $\sigma^2 = 1/(4n_V)$ , which lands *exactly* on the target constant  $2\sqrt{\log(2K)/(2n_V)}$  in expectation without a failure-probability term. I accepted it because it is both shorter and matches the stated bound’s constant exactly; I verified  $\sigma^2 = 1/(4n_V)$  from Hoeffding’s lemma myself.
2. **Decoupling B.3 from the stability bound B.2.** My first attempt at B.3 bounded the generalization gap of  $\tilde{w}_S$  using the B.2 stability constant directly, which produced an *extra*  $2G\sqrt{2\eta/(\lambda\alpha)}$  term — too large to match the claim. AI pointed out the cleaner decomposition: bound  $L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(w_S) + G\|\tilde{w}_S - w_S\|$  using Lipschitzness of the *true* risk, then apply the exact-RERM bound to  $w_S$ . This gives exactly the single  $G\sqrt{2\eta/(\lambda\alpha)}$  term. I accepted after re-deriving it.

### D.3 Suggestions I rejected or substantially modified

1. **Wrong strong-convexity constant.** AI at one point asserted that the regularized objective  $F_S = L_S + \lambda\Psi$  is “ $\alpha$ -strongly convex.” It is  $\lambda\alpha$ -strongly convex (the  $\lambda$  multiplies  $\Psi$ ). This matters in B.1: the correct bound is  $\|\tilde{w}_S - w_S\| \leq \sqrt{2\eta/(\lambda\alpha)}$ , and dropping the  $\lambda$  would have given the wrong dependence. I corrected it.
2. **Off-by-constant in B.5.** A draft treated the constraint  $\|u\|_2 \leq B$  as  $\Psi(u) \leq B^2$ , but  $\Psi(u) = \frac{1}{2}\|u\|_2^2 \leq \frac{1}{2}B^2$ . I redid the  $\lambda$ -optimization with the correct  $\frac{1}{2}B^2$ , yielding  $\lambda = \frac{2R}{B\sqrt{n}}$  and the leading term  $\frac{2RB}{\sqrt{n}}$  (rather than an inflated constant).
3. **Over-claiming on the “price of tuning.”** AI initially described the adaptivity cost as “ $\sqrt{\log(B_{\max}/B_{\min})/n}$ .” That is the cost if  $K \propto B_{\max}/B_{\min}$ , but the dyadic grid has  $K = O(\log(B_{\max}/B_{\min}))$ , so the validation term is the *doubly*-logarithmic  $\sqrt{\log \log(B_{\max}/B_{\min})/n}$ . I corrected the write-up in A.4.

## D.4 How I verified correctness

Since the deliverables are proofs, verification was step-level:

- **Constant tracking.** For every boxed bound I checked that the target constant emerges, not just the rate: the  $2\sqrt{\log(2K)/(2n_V)}$  in A.1, the  $\frac{5}{2}\sqrt{2}$  in A.3 (from  $\max_{c \in [1/2, 2]}(c + 1/c) = 5/2$ ), the collapse  $\frac{5}{2}\sqrt{2} \cdot \sqrt{2} = 5$  in A.4, and  $(2\sqrt{2} + 1)$  in B.4.
- **Limit/sanity checks.** As  $\eta \rightarrow 0$ , B.1–B.5 reduce to the exact-RERM statements; B.4’s condition  $\eta = O(n^{-3/2})$  is below the statistical  $n^{-1/2}$  scale, as expected. In C.3, the optimal  $q_j \propto r_j/b_j$  I verified by plugging back in to confirm both Cauchy–Schwarz factors equal  $\sum_j b_j r_j$ .
- **Independence bookkeeping.** The crux of A.2 is that  $h_\lambda = A_\lambda(T)$  is independent of  $V$  given  $T$ ; I made sure the conditioning order (condition on  $T$ , apply A.1 over  $V$ , then  $\mathbb{E}_T$ ) was airtight before trusting the oracle inequality.

## D.5 Updates to my AI workflow (5 most recent)

1. **Added a “constant audit” pass to my proof checklist.** After the B.3/B.5 constant slips, I now make a final pass that re-derives every boxed constant independently of the AI’s text. *Why:* AI errors in these problems are almost always constant/factor errors, not structural ones.
2. **Added a “strong-convexity parameter” reminder to CLAUDE.md’s notes section.** A standing note: regularized objectives carry the  $\lambda$  into the strong-convexity modulus ( $\lambda\alpha$ , not  $\alpha$ ). *Why:* this exact mistake recurred and is easy to miss.
3. **Standardized the train/validation split notation prompt.** I now state  $n_T, n_V$  (and which sample each quantity uses) explicitly in the prompt before asking for help. *Why:* AI mixed up  $n_T$  and  $n_V$  in an early A.2 draft.
4. **“Reduce to known limit” verification habit.** I added a step that checks every approximate/perturbed bound against its exact ( $\eta \rightarrow 0$ ) special case. *Why:* it instantly caught the spurious extra stability term in B.3.
5. **Kept the `build_pdfs.sh` pandoc pipeline from HW7 unchanged** (theory + ai\_report  $\rightarrow$  combined hw8\_writeup.pdf), since it has been reliable. *Why:* a stable, scripted build removes a class of last-minute formatting errors before submission.

I used AI for proof-checking, algebra verification, and exposition only; the proof strategies and all final corrections are my own.